

Driving the Hyde Park BESS cost down: \$57M to \$43M, and who owns each dollar

Sixteen levers, three gates and a design-to-cost target that take Project 21334 from the \$57.2M approval level to a \$38.5 to 42.9M re-scoped project, with the decisions leadership is asked to make now

Cost Estimating Center of Excellence | Leadership brief | Basis: ESF Total Rev 3, 24 Sep 2026 | 5 Oct 2026

Internal. Contains vendor-confidential pricing. Backup: Hyde Park BESS Cost Analysis Deck (41 slides).

WHAT CAN BE DONE

The number can be driven from \$57.2M to \$38.5 to 42.9M. About a third of that needs only a decision.

\$57.2M

Approved, Rev 3

24 Sep 2026; includes \$7.6M of risk and contingency and \$20.3M of loaders

\$50.7 to 52.6M

Contained: decisions only

Six decisions, no scope change, no study, by the November 2026 PAF

\$38.5 to 42.9M

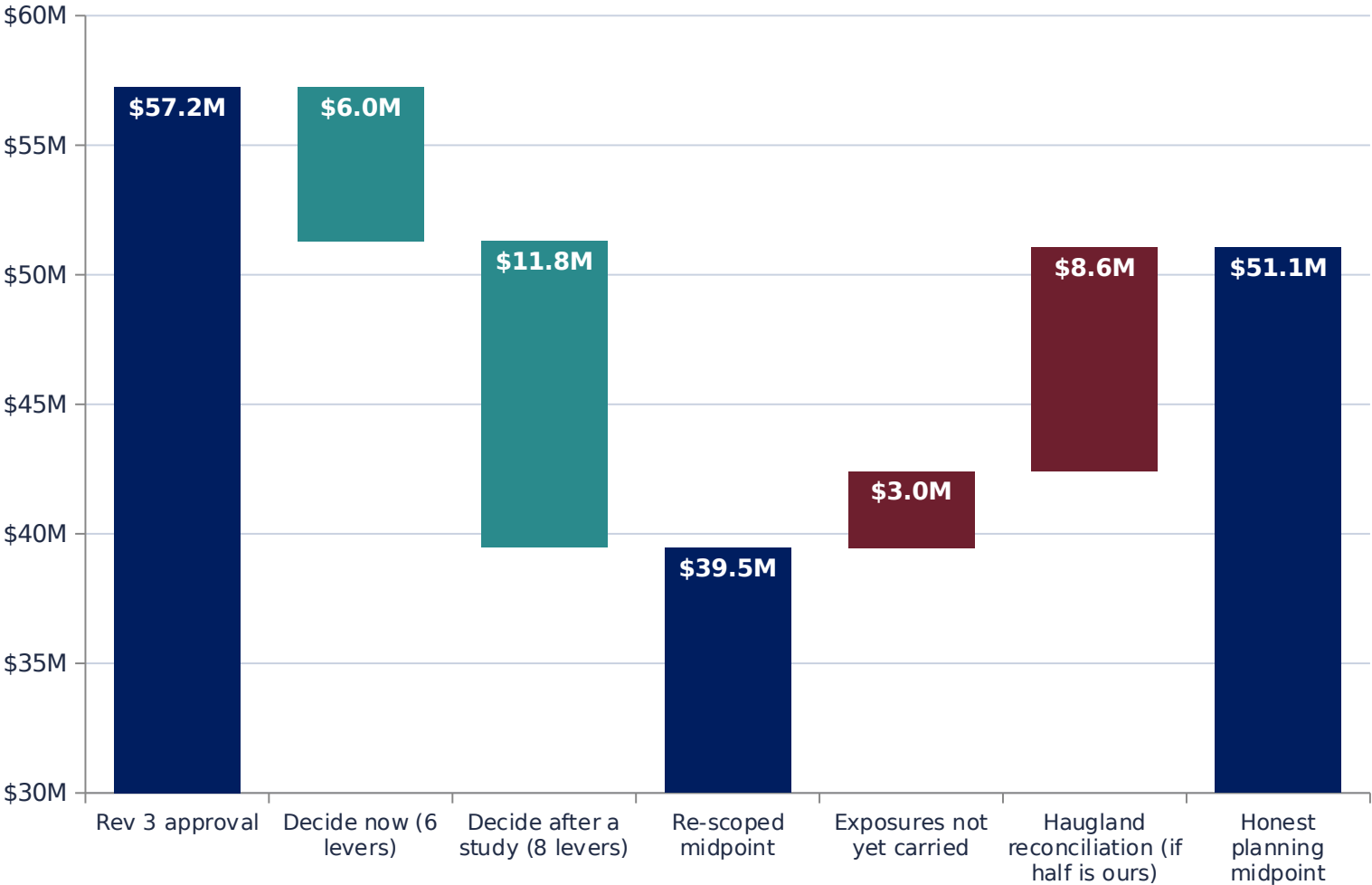
Re-scoped after four studies

Geotech, hosting capacity, the Station 496 protection basis, the Haugland scope map; Q1 2027

- **Decide now (no study):** \$4.9 to 7.0M loaded. Strike the escalation on a fixed price, re-word two risk lines that duplicate it, rule on the overhead basis, strike the double-counted disconnects, pay the battery at delivery.
- **Decide after a study (Q1 2027):** \$10.6 to 13.1M loaded. Engineering's proposal (Station 496 to RTU, RTAC and transfer trip only; two feeders instead of four), civil sized after geotech, a crew build-up for install.
- **Outside the estimate, larger than both:** \$11.2 to 16.5M of Section 48E credit on the revenue requirement, plus \$2.8 to 4.1M if the landfill qualifies as an energy community. Owner: Tax.
- **Then hold it:** a station-direct design-to-cost target of \$1,250 per kWh (\$50M loaded), competitive lump-sum pricing before construction, and a monthly number someone owns.

THE LEVERS, GROUPED BY WHAT THEY NEED

Where the dollars come from: a bridge from the approval level to the re-scoped project, loaded



Group	Loaded
Decide now, 6 levers	\$4.9 to 7.0M
After a study, 8 levers	\$10.6 to 13.1M
Subtotal, the estimate	\$15.4 to 20.1M
Exposures to add back	-\$2.0 to 3.9M
Haugland gap, direct	-\$4 to 8M
48E and adder, revenue req.	\$11.2 to 16.5M

Midpoints plotted. Every lever is sized from the Rev 3 line items and loaded at the marginal loader of 1.44 (contingency, basis-driven indirects and AFUDC scale with direct). The two red steps are not levers: they are what must be managed so the savings stick (slide 9).

GROUP 1: DECISIONS, NO STUDY

Decide now, before the November PAF: six levers worth \$4.9 to 7.0M, none of which changes scope

#	Lever	Loaded	Odds	What it takes	Owner	By
A1	Strike the escalation carried on the fixed-price battery	\$1.6M	80%	Decision; Exhibit C read for pass-through	Cost Estimating / Sourcing	Nov 2026
A2	Re-word two risk lines that duplicate that escalation	\$1.3 to 1.8M	70%	Re-word and re-size the two lines	Cost Estimating + PM	Nov 2026
A4	Take the E&S overhead basis off the vendor contract	\$1.0 to 2.1M	30%	Accounting ruling	Accounting / Cost Estimating	Nov 2026
A5	Split commissioning hours with Nomad in the test plan	\$0.2 to 0.5M	50%	Test plan split written	Capital Projects Eng. + Nomad	Nov 2026
B5	Strike the disconnects and PTs the Siemens skids already carry	\$0.4M	90%	Siemens confirms skid scope in writing	Substation Eng. + Cost Estimating	Nov 2026
C3	Pay the battery at delivery, not a year ahead (AFUDC)	\$0.3 to 0.7M	50%	Nomad milestone schedule in the SOV	PM + Sourcing + Treasury	Nov 2026
Total		\$4.9 to 7.0M				

■ **The ask:** approve these six as estimate corrections for the November 2026 PAF. The odds are the simulation's assessed probability of each landing; the expected value of the six is about \$3.9M.

GROUP 2: DECISIONS THAT NEED A STUDY

Decide at Gate 1 (Q1 2027): eight levers worth \$10.6 to 13.1M, each closed by a study already on the critical path

#	Lever	Loaded	Odds	What closes it	Owner	By
A3	Confirm the four design-basis items Haugland priced as deducts	\$0.9 to 1.2M	with T1	Haugland-to-Rev 3 scope map, signed	Substation Engineering	Q1 2027
B1	Station 496: relay program out; RTU, RTAC and transfer trip only	\$6.1 to 6.6M	60%	Protection-basis memo; RTU/RTAC scope priced	P&C Engineering + System Protection	Dec 2026
B2	Two feeders instead of four: two skids, two metering sets, two duct-bank runs	\$1.5 to 1.7M	60%	Hosting-capacity study at 5 MW per feeder; N-1 accepted	System Planning + Substation Eng.	Q1 2027
B3	Size the platform and wall after geotech, not before	\$0.4 to 1.2M	40%	Geotech report; MassDEP post-closure conditions	Civil + Capital Projects Eng.	Q1 2027
B4	Battery install labor from a crew build-up	\$0.7 to 1.1M	60%	Crew build-up with Nomad and HEG	Cost Estimating + Nomad + HEG	Q1 2027
B6	GSU install hours and count settled in writing	\$0.3 to 0.3M	60%	Topology and GSU count in writing	Substation Eng. + Nomad	Q1 2027
B7	Lighting to the security need, not 12 HPS luminaires	\$0.2 to 0.3M	70%	Lighting design	Substation Eng.	Q1 2027
B8	Siting and PM task budgets from outside counsel	\$0.5 to 0.8M	40%	Task budget from outside counsel	Siting + PM	Q1 2027

Total

\$10.6 to 13.1M

■ **The ask:** fund and schedule the four closers now (geotech, hosting capacity at 5 MW per feeder, the Station 496 protection-basis memo, the Haugland scope map) so every one of these is decided before IFC, not discovered after.

GROUP 2, THE TWO LARGEST LEVERS, ON ONE MEMO AND ONE STUDY

Engineering's 5 Oct proposal is worth \$7.6 to 8.3M: Station 496 to RTU, RTAC and transfer trip only, and two feeders

\$6.1 to 6.6M

Station 496, net

\$7.25M relay program out; \$0.7 to 1.1M kept for the RTU, RTAC and transfer trip, which Rev 2 names but does not price

\$1.5 to 1.7M

Two feeders, loaded

Two breaker skids, two metering sets, two duct-bank runs, two riser-pole pairs; four GSUs unchanged

\$42.4 to 45.0M

Contained case plus this

The six no-study decisions and the proposal; nothing else in the register

What makes it real	Owner	By	What it decides
Protection-basis memo: the existing SEL-551 and REF610 relays plus an RTAC transfer trip over the dark fiber meet the interconnection requirement	System Protection	Dec 2026	Without it the \$7.25M stays
Price the RTU, RTAC/Axion, panel, fiber terminations and point lists (named in Rev 2, no line)	P&C Eng. + Cost Estimating	Nov 2026	Allowance of \$0.30 to 0.55M direct until then
Asset-condition check on the legacy relays	Asset Management	Dec 2026	If they must go anyway, the cost moves to that program; customers do not save it
Hosting-capacity and protection study at 5 MW per feeder; N-1 position accepted in writing	System Planning	Q1 2027	Two feeders, and whether voltage mitigation is needed (\$0.2M to \$0.5M exposure)
D-Line 24211 re-scoped to two tie-ins	Distribution Eng.	Q1 2027	Outside 21334; the 2023 \$4.8M memo shrinks too

■ **The honest reading:** the largest single lever in the register and about a quarter of the simulated variance on its own. It is a saving to customers only if the relays were BESS-driven scope; say which before it goes in the PAF.

GROUP 3: REVENUE REQUIREMENT

The largest lever is not in the estimate: \$8.4 to 12.4M of Section 48E credit to customers, and nobody owns it yet

\$8.4 to 12.4M

30% credit, eligible basis

Battery, install, balance of plant (\$28M); up to \$41M with station, site and testing

+\$2.8 to 4.1M

Energy-community adder

Only if the closed landfill qualifies as a brownfield; not assumed

65%

Non-PFE cost ratio

Cell origin on the Voyager units decides whether the credit is zero

Decision	Owner	By	Why it matters
Eligible-basis opinion: which layers are energy storage property	Tax	Dec 2026	Sets the \$28M to \$41M range
FEOC cost-ratio test on the Nomad cells (Notice 2026-15)	Tax + Sourcing + Nomad	Before final drawings	A failed test is a zero credit
Normalization election; DPU treatment of the credit	Regulatory	2027 filing	Decides how fast customers see it
Prevailing wage and apprenticeship in the construction contracts	Sourcing + Construction	Gate 3	6% credit becomes 30%

■ **The ask:** open the 48E memo with Tax this month. It is larger than every estimate lever combined and it has a construction-start deadline.

Hold it: a station-direct target of \$1,250 per kWh, \$50M loaded, with change control and a monthly number

\$1,250/kWh

Station-direct target

\$25M direct: the reference-class 75th percentile for a built project this size

\$1,483/kWh

Where Rev 3 sits

\$29.7M direct, the 77th percentile of 35 built projects at or below 100 MWh

\$50M

Loaded target

The contained case, held through IFC, construction pricing and in-service

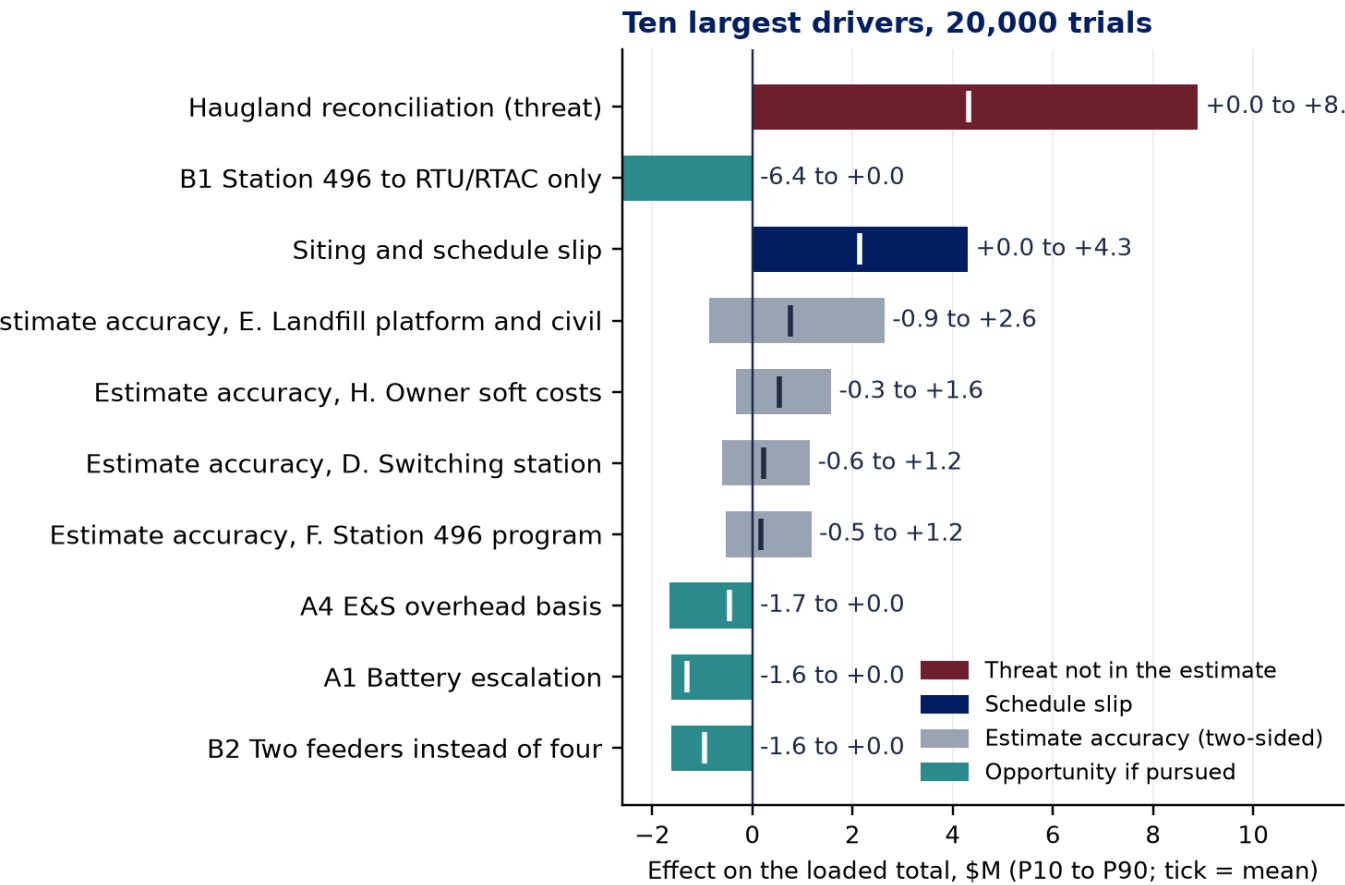
- **Why a target and not a forecast:** an estimate that is re-forecast every revision has moved from \$43.9M to \$57.2M in three years. A target with change control turns it into a managed number.
- **The rule:** no scope is added without a funded source. A change control board with the CoE decides every addition against the target, in writing.
- **The measure:** station-direct \$/kWh against target, forecast at completion, open scope questions, the exposures register. One page, monthly, to the sponsor, until in-service.
- **Why this level:** it is defensible outside the company. A regulator can be told the project was designed to the 75th percentile of what utilities have actually built, and why it stopped where it stopped.

Buy it differently: competitive lump-sum pricing before construction, owner-furnished equipment, a change-order cap

Action	What it changes	Owner	By
Competitive lump-sum pricing for station, civil and install: Haugland plus two bidders	Converts a unit-library estimate at 30% engineering into a bid; retires the +25% construction stress test	Sourcing + PM	Q3 2027
Owner-furnished GSUs and breaker skids on alliance pricing	Takes contractor markup off \$3.8M of equipment; locks the 29 to 70 week transformer lead time	Sourcing + Substation Eng.	Q2 2027
Nomad payment milestones aligned to the January 2028 delivery	Removes a year of AFUDC on \$12M; protects the siting float	PM + Sourcing + Treasury	Nov 2026
Change-order cap and a tariff pass-through clause read before signature	Bounds the \$0.6M Nomad change-order exposure in the simulation	Sourcing + Legal	Nov 2026
Outage windows at Station 496 priced explicitly for whatever work remains	Unpriced today; small if the relay program goes	P&C Eng. + EMA	Gate 1

■ **The ask:** approve the contracting strategy at Gate 3 rather than defaulting to the alliance contractor on unit rates. The simulation's two largest accuracy ranges, the civil layer and Station 496, are both unit-library numbers with no bid behind them.

Two risks can eat every dollar saved: the Haugland reconciliation and the siting calendar. Both have an owner and a date.



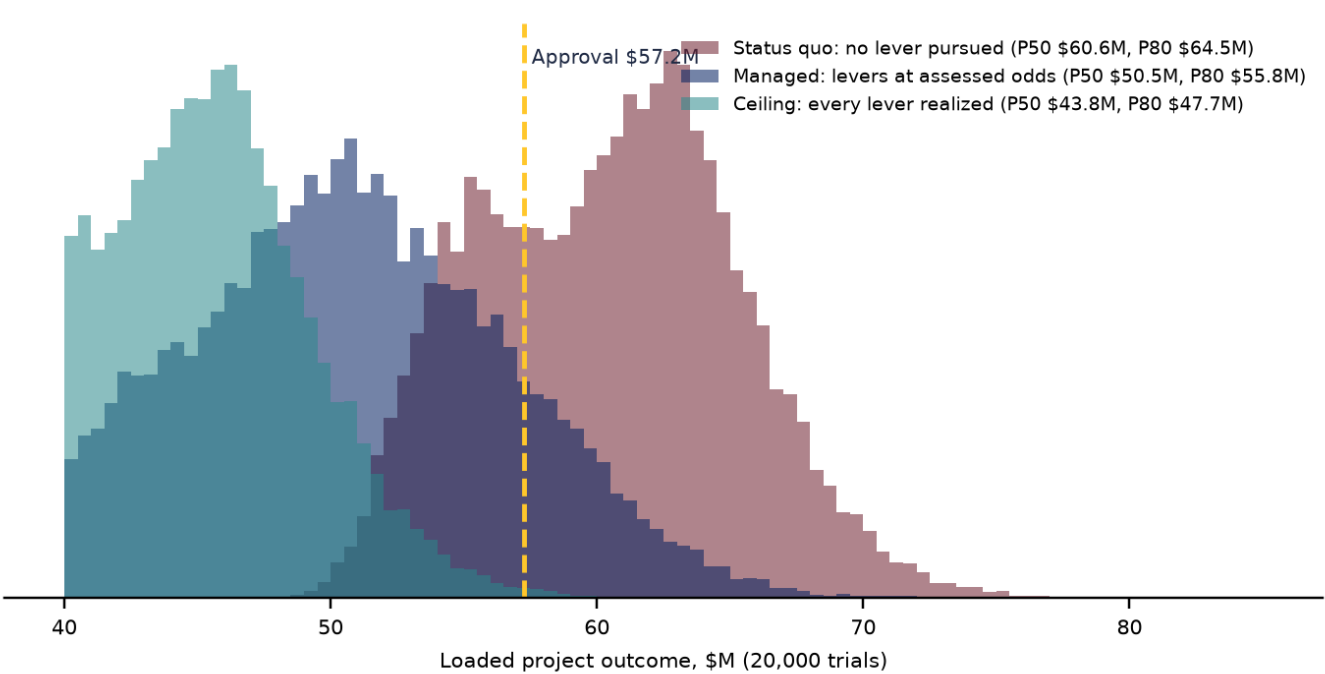
- **Haugland reconciliation, +\$4 to 8M direct, 43% of the variance.** Haugland's Class 3 base for overlapping scope is \$19.2M; Rev 3 carries \$8.3M. Until a signed scope map says which is right, every saving on slides 4 and 5 is provisional. Owner: PM + Cost Estimating + HEG. By: before the November PAF.
- **Siting schedule, +5 months at P50, +9 at P80, 11% of the variance.** Every month costs about \$0.37M in AFUDC, escalation and PM. Action: file Article 80 early, pre-brief the abutters on screening and sound, set the ZBA date within 90 days of filing. Owner: Siting + PM. By: Q1 2027.
- **Station 496 is now the second driver, 26% of the variance.** A \$6M swing that turns on one protection-basis memo; until it is signed the number has two modes.
- **Exposures to carry honestly, \$2.0 to 3.9M.** Sound mitigation (\$467K dropped from the 2023 estimate), visual screening, police details, NFPA 855 changes, the GSU count. Price them now; a surprise in 2028 costs more than a line today.
- **Everything on the left is ours to take.** No single lever exceeds \$2M; together they are worth about \$9.9M at the assessed odds and \$16.8M if every one lands.

The plan on one page: five gates, each with an owner, a date and the dollars it decides

When	What	Who	What it decides
Now to the Nov 2026 PAF	The six no-study decisions (A1, A2, A4, A5, B5, C3); sound mitigation and screening priced or registered; the 48E memo opened with Tax	Cost Estimating, PM, Accounting, Tax	Estimate hygiene, about \$3M to \$5M loaded; the exposures carried honestly
Gate 1, design basis (Q1 2027)	Geotech report; NFPA 855 hazard mitigation analysis; hosting-capacity and protection study at 5 MW per feeder; Station 496 protection-basis memo; GSU count and topology in writing; Haugland-to-Rev 3 scope map signed	Substation Eng., Civil, System Planning, System Protection, HEG	The Station 496 case (\$6.1 to 6.6M), the two-feeder case (\$1.5 to 1.7M), the civil layer, the GSU exposure, the Haugland gap
Gate 2, design-to-cost (before IFC, mid-2027)	Station-direct target of \$1,250 per kWh and \$50M loaded; monthly trend review; change control board with the CoE; no scope added without a funded source	PM (accountable), Cost Estimating, sponsor	Turns the estimate into a managed number instead of a forecast
Gate 3, commercial strategy (Q3 2027)	Competitive lump-sum pricing for station, civil and install (Haugland plus two); owner-furnished GSUs and skids; Nomad milestones aligned to delivery; change-order cap; Station 496 outage windows priced	Sourcing, PM, Construction	Converts unit-library construction into a bid; retires the +25% construction stress test
Standing, monthly to in-service	One-page cost-containment report: station-direct \$/kWh against target, forecast at completion, open scope questions, exposures register, the 48E and FEOC milestones	PM, CoE	What leadership sees every month

Target: \$1,250 per kWh station-direct (\$25M direct, \$50M loaded); the CoE reports the trend monthly from the November 2026 PAF.

What the plan is worth: \$10.1M at the median if the levers are worked at their odds, \$16.8M if every one lands



Posture	P50	P80	Over \$57.2M
Nothing pursued	\$60.6M	\$64.5M	70%
Levers at their odds	\$50.5M	\$55.8M	14%
Every lever lands	\$43.8M	\$47.7M	0%

- **Read it plainly:** with nothing pursued, 70% of trials end above the approval level, because the Haugland gap, the unpriced exposures and the schedule are real and the estimate carries none of them.
- **With the plan:** the median lands at \$50.5M and 47% of trials come in under \$50M.
- **The reserve:** \$7.6M of risk and contingency covers the modeled ranges at about P75 only if the plan is executed.

\$57M becomes \$38.5 to 42.9M through sixteen named levers, three gates and one target. Six of them need only a yes.

- **1. Approve the six no-study corrections for the November 2026 PAF** (\$4.9 to 7.0M loaded).
Owners: Cost Estimating, PM, Accounting, Sourcing.
- **2. Fund the four closers now** (geotech, hosting capacity at 5 MW per feeder, the Station 496 protection-basis memo, the Haugland scope map) so the eight study-gated levers (\$10.6 to 13.1M), led by Engineering's Station 496 and two-feeder proposal, are decided before IFC.
- **3. Open the Section 48E memo with Tax this month** (\$8.4 to 12.4M on the revenue requirement; FEOC cell-origin test on the Nomad units).
- **4. Adopt the design-to-cost target** (\$1,250 per kWh station-direct, \$50M loaded) with change control and a monthly report from the CoE.
- **5. Approve competitive lump-sum pricing of the station, civil and install scope at Gate 3**, with owner-furnished GSUs and skids.
- **6. Assign the two risks that can undo all of it:** the Haugland reconciliation (PM + Cost Estimating + HEG, before the PAF) and the siting calendar (Siting + PM, Q1 2027).

APPENDIX: SAVINGS REGISTER, \$15.1 TO 19.5M ON THE ESTIMATE, \$11.6 TO 17.2M ON THE REVENUE REQUIREMENT

Backup: the sixteen levers as sized from the Rev 3 line items

#	Lever	Layer	Direct	Loaded	Conf.	Owner
A1	Remove escalation carried on the fixed-price Nomad balance	A	\$1.1M	\$1.6M	High	Cost Estimating / Sourcing
A2	Risk register: labor-cost and material-cost lines overlap the escalation already in the directs	J	\$1.3 to 1.8M	\$1.3 to 1.8M	Medium	Cost Estimating + PM
A3	Confirm the four written design-basis items Haugland priced as deducts	D/E	\$0.6 to 0.8M	\$0.9 to 1.2M	Medium	Substation Engineering
A4	Engineering & supervision overhead basis applied to the vendor turnkey contract	L	\$1.0 to 2.1M	\$1.0 to 2.1M	Low	Accounting / Cost Estimating
A5	Commissioning hours carried from E-23-373 overlap Nomad's commissioning support	G	\$0.1 to 0.3M	\$0.2 to 0.5M	Low	Capital Projects Eng. + Nomad
B1	Station 496: relay program out; RTU, RTAC and transfer trip only (Engineering, 5 Oct 2026)	F	\$6.1 to 6.6M	\$6.1 to 6.6M	Medium	P&C Engineering + System Protection
B2	Two feeders instead of four: two skids, two metering sets, two duct-bank runs (Engineering, 5 Oct 2026)	C/D	\$1.0 to 1.2M	\$1.5 to 1.7M	Medium	System Planning + Substation Eng.
B3	Platform and retaining wall sized after geotech, not before	E	\$0.3 to 0.8M	\$0.4 to 1.2M	Low	Civil + Capital Projects Eng.
B4	Battery installation labor from a crew build-up instead of 25% of a 2023 total; fire suppression bought twice?	B	\$0.5 to 0.7M	\$0.7 to 1.1M	Medium	Cost Estimating + Nomad + HEG
B5	Disconnect switches and PT cutouts priced as separate lines although the Siemens skids carry them	D	\$0.3M	\$0.4M	High	Substation Eng. + Cost Estimating
B6	GSU installation hours: 524 man-hours per pad-mount; count to be settled (four carried, twelve mentioned)	C	\$0.2 to 0.2M	\$0.3 to 0.3M	Medium	Substation Eng. + Nomad
B7	Switchyard lighting: 12 HPS luminaires and \$200 a foot of hookup on an unmanned one-acre pad	D	\$0.2M	\$0.2 to 0.3M	Medium	Substation Eng.
B8	Siting and project management: external siting at \$1.05M and a \$157K siting line inside Station 496	H/F	\$0.4 to 0.6M	\$0.5 to 0.8M	Low	Siting + PM
C1	Section 48E investment tax credit on the storage property	rev. req.	\$8.4 to 12.4M	\$8.4 to 12.4M	Medium	Tax + Regulatory
C2	Energy-community adder if the closed landfill qualifies as a brownfield	rev. req.	\$2.8 to 4.1M	\$2.8 to 4.1M	Low	Tax + Environmental
C3	AFUDC: pay the battery at delivery, not a year ahead; protect the siting float	M	\$0.3 to 0.7M	\$0.3 to 0.7M	Medium	PM + Sourcing + Treasury

Full basis, evidence lines and what must close for each lever: Hyde Park BESS Cost Analysis Deck slides 27 to 36, Whitepaper section 8, Workbook tabs Savings Register and Scope Challenge.